

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Market-Led Partnerships for Resilient Agri-Food Systems in the Terai Region of Nepal

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.1 · Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

This solution track aims to scale bundled agri-food innovations in Nepal's Terai region through market-led public-private-community partnerships. It focuses on transforming subsistence smallholders into commercial farmers by improving access to and adoption of proven technologies in seeds, soil fertility, irrigation, and mechanization, supported by inclusive multi-stakeholder platforms and stronger market linkages.

The solution track builds on 4+ years of successful piloting led by CIMMYT and IWMI under the Maize Commercial Model (MCM), which has already driven a 700% increase in spring maize adoption and significantly raised yields and farmer incomes. With catalytic funding from the Scaling for Impact (S4I) program, the initiative now expands from Western to Eastern Nepal, targeting systemic change in how agricultural inputs are delivered, financed, and governed.

Core Partners: CIMMYT, IWMI **Key Collaborators:** Farmer cooperatives, agro-dealers, seed companies, feed mills, local/provincial governments, irrigation agencies, industry associations, universities, and bilateral projects.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **Quality seeds** (hybrids, OPVs, local landraces)
- **Integrated Soil Fertility Management** (4R nutrient stewardship, composting, green manure)
- **Irrigation and water management** (scheduling, efficient application, infrastructure)
- **Mechanization** (seed drills, planters, combined harvesters)
- **Multi-stakeholder platforms (MSPs)** for co-design, coordination, and market linkage

2.2 Evidence of Impact

- 20–40% yield increase in rice, legumes, vegetables; ~100% increase in maize
- Sowing costs reduced by 50% through mechanization
- Spring maize area expanded from pilot to 2,000+ hectares (2021–2025)
- Maize sales via cooperatives grew 850% to USD 3.3M
- Farmgate prices rose 81% due to better market coordination

2.3 Scaling Targets

Households reached	58,300 (39,300 rice + 19,000 maize)	237,500 HH (1.2M people, 40% women/youth)	
Area under improved practices (ha)	50,000 ha	257,730 ha across 11 districts	
Irrigated area expansion (ha)	2,000 ha	10,000 ha	
Jobs supported	3,000	10,000+ in agri-inputs sector	
Investment leveraged	USD 10M	USD 30M+	

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The track adopts a **market-systems approach**, moving away from donor-driven input distribution toward **sustainable commercial models**. Key elements:

- **Demand-driven input roadmaps** – developed with MSPs to match supply with real-time needs.
- **Public-private-community partnerships (PPCP)** – formal and informal collaborations among local governments, private suppliers, cooperatives, and buyers.
- **Last-mile delivery** via trained agro-dealers, lead farmers, digital advisories, and local colleges.
- **Output market linkages** – assured buyback agreements with processors/feed mills to create pull for quality inputs.

3.2 Theory of Change

If smallholders gain access to bundled innovations through market-led partnerships, and input and output markets are better coordinated through inclusive MSPs, then farm productivity, profitability, and resilience will increase, leading to sustainable local supply chains, private sector investment, and policy uptake, ultimately contributing to food security, rural employment, and climate-resilient agri-food systems in Nepal.

3.3 Key Activities Funded by S4I

- **Mobilization:** Stakeholder mapping, MSP facilitation, cropping calendars, training of trainers.
- **Input market development:** Delivery system design, actor training, linkage with output markets.
- **Partnership and governance:** Joint strategy design, market assessments, investment forums, policy dialogues.

4. Partners and Roles

Demand Partners	14 seed producer-cooperatives, 35 farmer cooperatives, 28 agro-dealers, 28 MSMEs/feed and food mills, 7 seed companies, Feed mills and food processing industries	Create market demand for produce, provide buyback agreements, signal quality requirements, drive commercial incentives for adoption
Scaling Partners	21 local governments, 3 provincial governments (Koshi, Lumbini, Sudurpaschim), Ministry of Agriculture and Livestock Development (MoALD), Department of Water Resources and Irrigation (DWRI), 4 Water User Association Groups (Babai and Rani-Jamara-Kulariya irrigation systems), Nepal Seed Entrepreneurs' Association (SEAN), Nepal Agriculture Machinery Association (NAMEA), confederation of Nepalese Industries (CNI), NGOs and development partners, Agricultural universities and colleges, bilateral programs (MoALD's Rice Mill Model, JICA seed project, provincial schemes)	Co-design roadmaps, align policies and budgets, provide subsidies and water access, monitor, facilitate coordination, mobilize funding and extension support

4.4 Funding Model: S4I does **not fund input purchases**. Instead, it enables system coordination, capacity building, and evidence generation. Inputs are financed by farmers, cooperatives, private sector, and government subsidies.

5. Enabling Environment and Risks

5.1 Opportunities

- Strong government prioritization of agri-inputs in provincial and federal budgets.
- Existing irrigation infrastructure (e.g., Babai, Rani-Jamara-Kulariya systems) with potential for modernization.
- Growing private sector interest in local sourcing and value chain development.
- Digital tools available (e.g., National Irrigation and Infrastructure Monitoring System – NIIMS).

5.2 Challenges and Mitigation

- **Complexity of bundle:** Focus will be narrowed to **seed, soil, and irrigation** initially; mechanization may be streamlined.
- **Partner capacity:** Scaling scan and capacity analysis will be conducted to ensure partners can grow with the program.
- **Sustained financing:** Roadmaps and investment forums will help mobilize follow-on funding from government and private sector.

- **Gender and inclusion risks:** Specific targeting of women and youth as lead farmers and entrepreneurs will be built into activity design and monitored.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process outcomes:** # of functional MSPs, PPP contracts signed, roadmaps developed.
- **Behavior change:** Shift in input purchasing patterns, adoption of bundled practices.
- **System outcomes:** Increase in private sector stockholding, reduction in subsidy dependency, improved market transparency.

6.2 Adaptive Management

- Quarterly check-ins with S4I team.
- Peer learning via proposed **Teams chat for all solution tracks**.
- Mid-year review to refine targets and activities based on evidence.

7. Next Steps and Timeline (2026)

Q1	Finalize target districts and partners; conduct capacity assessment; launch MSPs
Q2	Input roadmap development; training of trainers; cropping calendar dissemination
Q3	Input delivery monitoring; output market linkages; field days
Q4	Harvest assessments; investment forum; annual review and planning for 2027

Communications and Knowledge: Success stories, policy briefs, and scaling lessons will be shared at key cropping stages and through CGIAR channels.

8. Contact

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